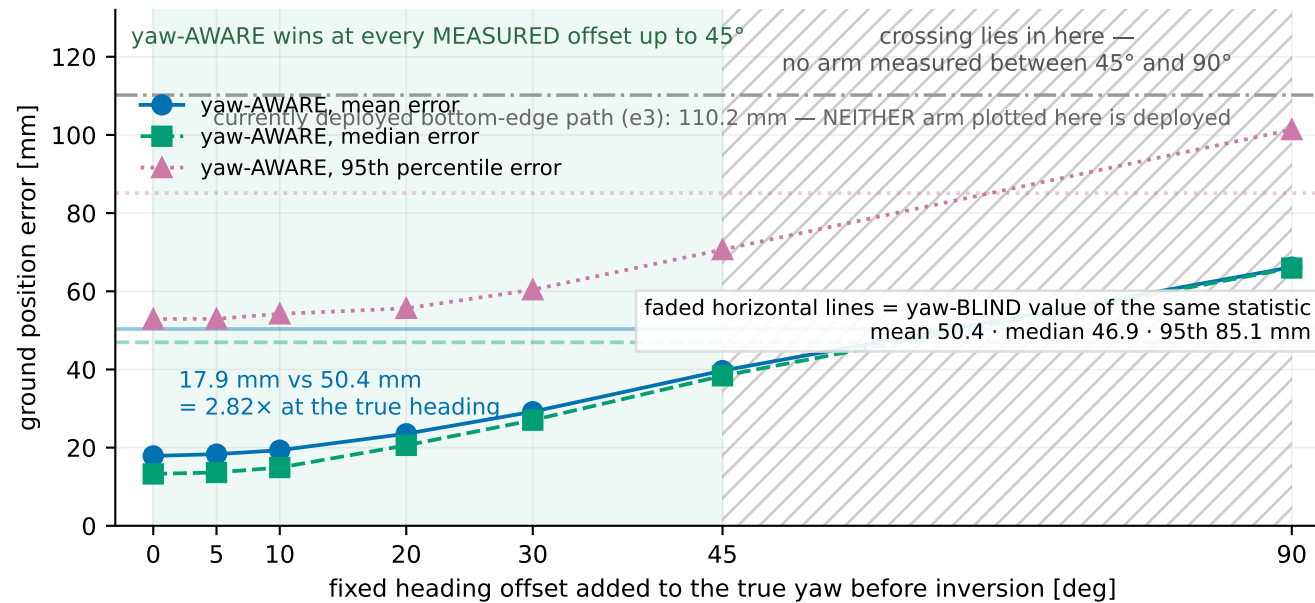
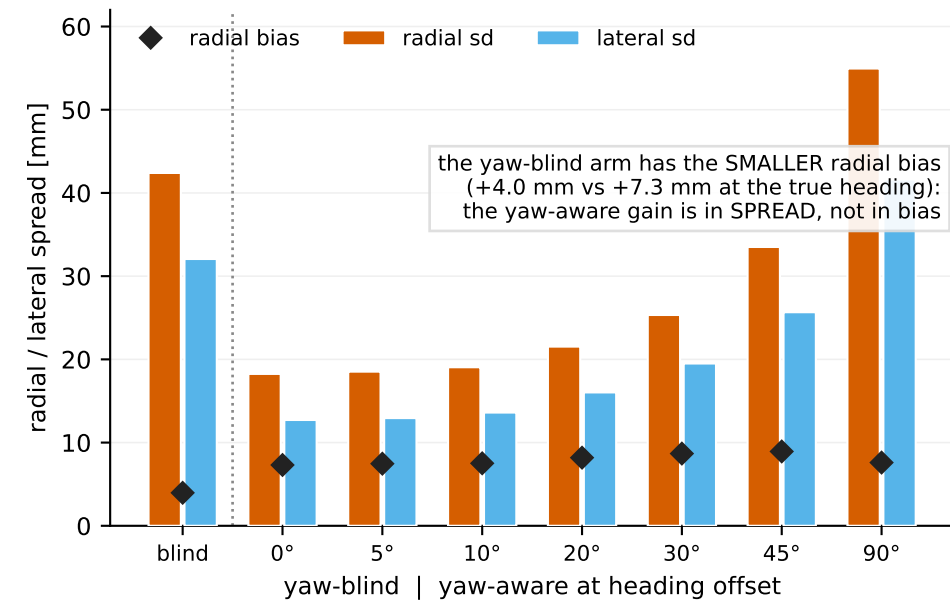


Yaw-AWARE beats the yaw-BLIND plane inversion — 50.4 mm → 17.9 mm — and still wins at 45° of heading error

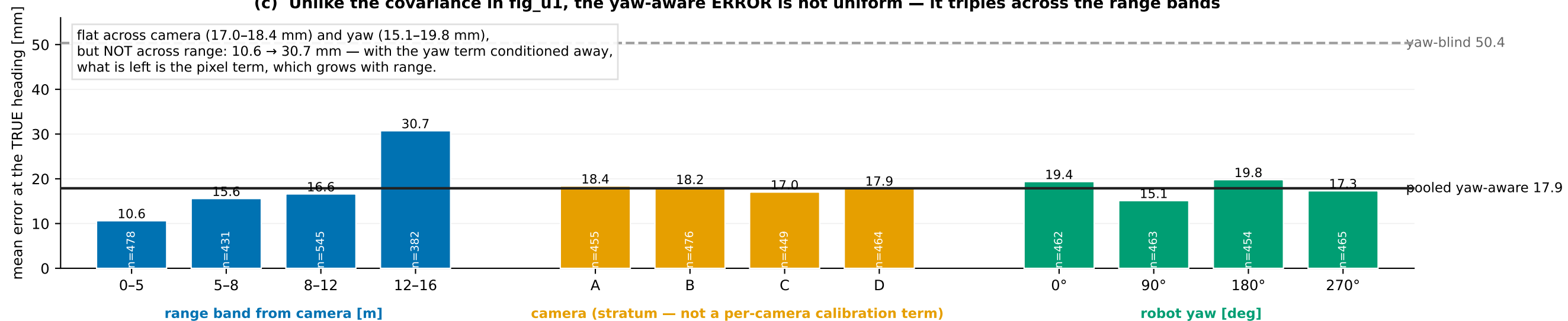
(a) Conditioning on heading is worth 2.8×, and it degrades gracefully: at 45° of heading offset it is still 21 % better than the yaw-blind inversion



(b) Both spreads inflate with heading offset while the bias stays put — the failure mode is variance, not offset



(c) Unlike the covariance in fig_u1, the yaw-aware ERROR is not uniform — it triples across the range bands



Markers sit on measured arms; straight segments between them are reading guides, not a fit, and nothing is extrapolated. The yaw-aware arm still wins at the largest offset measured below 90°, so the break-even is bracketed by the 45° and 90° arms rather than located at 45°. Range bands in (c) cover 1836 of the 1844 detections; 8 lie beyond 16 m.

Heading is the recorded true yaw plus a FIXED offset — a systematic offset, not zero-mean heading noise — so the yaw-aware arm is measured headroom, not a deployable estimator (it needs an iterative solve, a heading-quality monitor and a fallback). Ground truth is evaluation-only. 1844 detections, 4 discrete true yaws, all occlusion_state == clear; occlusion untested. Both arms are experiment-local; neither is a deployed runtime interface.